

Mobile Application Development

NoirList

Phase 1 — Team Task Split, Screens & Full UI/Theme Specification

Project Name: NoirList — a minimalist black & white To-Do List app. (Placeholder name — rename freely, just keep it consistent across the app title, splash screen, and report.)

#	Name	Student ID	Owns (Phase 1)
1	Mostafa Mohamed (Leader)	202202211	Splash + Login screens
2	Ahmed Sameh Ezzat	202202151	Signup + Home screens
3	Wafaa Abdulrazek	202202056	Add/Edit screen + Bonus + Report

Phase 1 Screens (Updated)

Login and Signup screens are added to Phase 1. Since Firebase Authentication only arrives in Phase 2, these two screens should work as local/mock auth for now: a simple form with validation that stores the user in memory (or local storage) and navigates to Home. Real backend auth gets wired in during Phase 2 without changing the UI.

- Splash Screen
- Login Screen (*new*)
- Signup Screen (*new*)
- Home Screen — task list, add/edit/delete, mark complete/incomplete
- Add/Edit Task Screen

Bonus Features Pulled Into Phase 1

Two bonus items are added now since they touch the UI foundation and are easier to build in from the start rather than retrofitted later:

- **Dark Mode** — toggle between the light (white/black) and dark (black/white-inverted) theme, saved with the rest of the app state.
- **Animations** — simple transitions: fade/slide on screen navigation, animated check on task completion, list item insert/remove animation.

Task Split — Collaborative Model

Instead of each member owning separate screens in isolation, everyone now contributes across all three phase steps — **Screens (UI)**, **Logic**, and **Testing** — so the whole team understands the full app. Work is still divided unevenly on purpose: Wafaa carries the lightest engineering load here since she owns the written report separately.

1. Screens (UI)

- **Mostafa:** Splash Screen (logo, app name, auto-navigation) and Login Screen UI (fields, buttons, error states).
- **Ahmed:** Signup Screen UI, Home Screen UI (AppBar, task list, empty state, FAB), Add/Edit Task Screen UI.
- **Wafaa:** Dark Mode toggle UI (switch placed in AppBar or a simple settings area).

2. Logic

- **Mostafa:** Login validation & local/mock authentication check; app navigation/routing setup (Splash → Login/Signup → Home).
- **Ahmed:** Signup validation logic; Task data model and full CRUD logic (add, edit, delete, mark Completed/Incomplete); state management binding the list to the UI.
- **Wafaa:** Light/Dark theme persistence logic (save and load the user's chosen theme).

3. Testing

- **Mostafa:** Test navigation flow across all five screens (Splash, Login, Signup, Home, Add/Edit) and fix broken transitions.
- **Ahmed:** Test CRUD operations and form validation edge cases (empty title, empty/incorrect login & signup fields).
- **Wafaa:** Final QA pass — confirm the app compiles and runs without errors in both light and dark mode.

Report (Wafaa, separate from the engineering split)

- Take screenshots of every screen (light + dark mode).
- Write the Phase 1 PDF report: Team Members, Application Description, Screenshots, Team Contributions.

Workload Summary

Member	Screens	Logic	Testing	Engineering Total
Mostafa Mohamed	2	2	1	5
Ahmed Sameh Ezzat	3	3	1	7
Wafaa Abdulrazek	1	1	1	3

Wafaa's engineering total (3) is the lowest, since she also owns the report and screenshots separately — touching all three phase steps without carrying the heaviest build load.

Team Leader Note

Mostafa (Team Leader) is additionally responsible for coordinating the final submission: zipping the project folder, building the APK, and merging everyone's work into the final deliverables.

Theme & Design System

Full black & white (monochrome) design system, including a dark mode variant for the bonus toggle.

Color Palette — Light Mode (default)

Token	Usage	Hex
Background	Screen backgrounds	#FFFFFF
Surface	Cards, sheets, dialogs	#FAFAFA
Primary text	Titles, main content	#000000
Secondary text	Subtitles, hints, completed tasks	#616161
Primary / accent	Buttons, FAB, active icons, checkmarks	#000000
Border / divider	Card borders, list dividers	#E0E0E0
Error	Validation errors (Login/Signup/Add-Edit)	#B00020
Disabled	Disabled buttons/fields	#BDBDBD

Color Palette — Dark Mode (bonus toggle)

Token	Usage	Hex
Background	Screen backgrounds	#000000
Surface	Cards, sheets, dialogs	#121212
Primary text	Titles, main content	#FFFFFF
Secondary text	Subtitles, hints, completed tasks	#B0B0B0
Primary / accent	Buttons, FAB, active icons, checkmarks	#FFFFFF
Border / divider	Card borders, list dividers	#2C2C2C
Error	Validation errors	#CF6679

Typography

- Font family: Poppins, Roboto, or Inter (any clean sans-serif, used consistently everywhere).
- App title / splash: 24–28sp, bold.
- Screen titles (AppBar): 18–20sp, semi-bold.
- Task titles: 15–16sp, regular (medium weight if unread/incomplete).
- Body / hints / helper text: 12–13sp, regular, secondary text color.

- Button text: 14–15sp, semi-bold, uppercase or title-case (pick one and stay consistent).

Spacing & Shape

- Base spacing unit: 8px (use multiples: 8 / 16 / 24 / 32 for padding and gaps).
- Screen padding: 16px horizontal minimum.
- Card / input border radius: 10–12px (soft, minimal — not fully rounded pill shapes).
- Button border radius: 10px.
- Card elevation: keep flat — use a 0.5–1px light-grey border instead of heavy drop shadows.

Component Styling

Buttons

- Primary button (Login, Sign Up, Save): filled black background, white text, 10px radius.
- Secondary button (Cancel, switch to Signup/Login): black text on transparent/white, or black outline.
- Disabled button: light-grey background (#BDBDBD), white/grey text.

Input Fields (Login, Signup, Add/Edit)

- White/surface background with a thin grey border (#E0E0E0); border turns black on focus.
- Black label and input text; grey placeholder/hint text.
- Error state: red border (#B00020) with a small red helper text below the field.

Task Card

- White surface, thin grey border, 12px radius.
- Circular checkbox outline; fills solid black with a white check when completed.
- Completed task title: grey + strikethrough. Pending task title: black, no strikethrough.
- Edit (pencil) and Delete (trash) icons, black outline style, right-aligned.

FAB & Icons

- FAB: solid black circle, white "+" icon, bottom-right on Home.
- Icon style: outline/line icons throughout (not filled), consistent stroke width, black or white depending on theme.

Animation Guidelines (Bonus)

- Screen transitions: fade or slide, 200–300ms duration, ease-in-out curve.
- Task complete toggle: quick scale/check animation on the checkbox (~150ms).
- Task add/remove: list item slide-and-fade in/out (~250ms) so the list never "jumps."
- Keep all animations short and subtle — the goal is polish, not flashiness.

Flutter ThemeData Reference

```
// Light theme
ThemeData(
  brightness: Brightness.light,
  scaffoldBackgroundColor: Colors.white,
  primaryColor: Colors.black,
  appBarTheme: const AppBarTheme(
    backgroundColor: Colors.white,
```

```

        foregroundColor: Colors.black,
        elevation: 0.5,
    ),
    floatingActionButtonTheme: const FloatingActionButtonThemeData(
        backgroundColor: Colors.black,
        foregroundColor: Colors.white,
    ),
    colorScheme: ColorScheme.light(
        primary: Colors.black,
        secondary: Colors.grey.shade800,
        error: const Color(0xFFB00020),
    ),
    fontFamily: 'Poppins',
)

// Dark theme (bonus toggle)
ThemeData(
    brightness: Brightness.dark,
    scaffoldBackgroundColor: Colors.black,
    primaryColor: Colors.white,
    appBarTheme: const AppBarTheme(
        backgroundColor: Colors.black,
        foregroundColor: Colors.white,
        elevation: 0.5,
    ),
    floatingActionButtonTheme: const FloatingActionButtonThemeData(
        backgroundColor: Colors.white,
        foregroundColor: Colors.black,
    ),
    colorScheme: ColorScheme.dark(
        primary: Colors.white,
        secondary: Colors.grey.shade300,
        error: const Color(0xFFCF6679),
    ),
    fontFamily: 'Poppins',
)

```